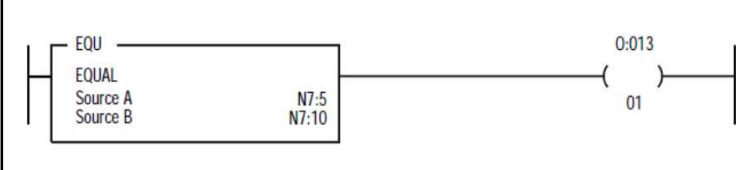
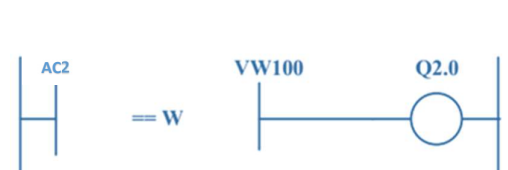
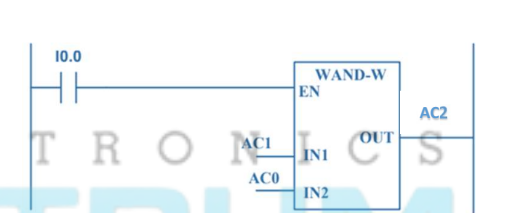
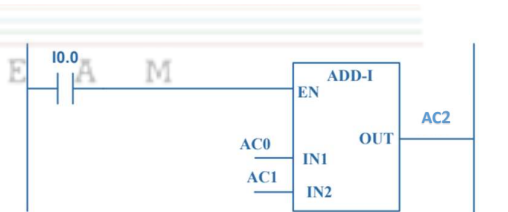
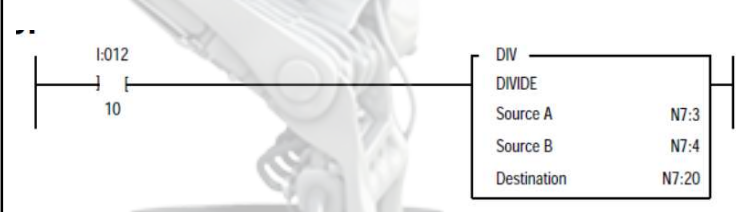
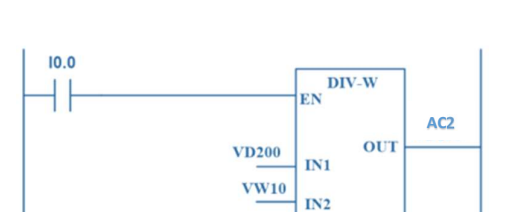
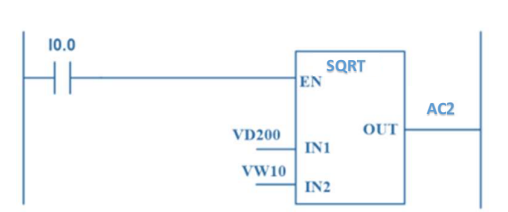
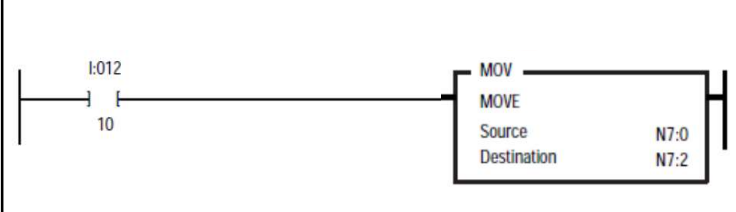
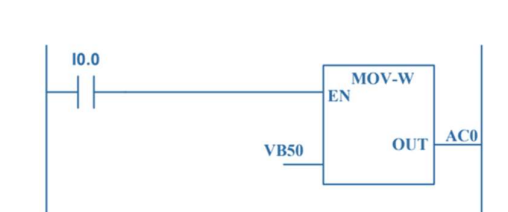
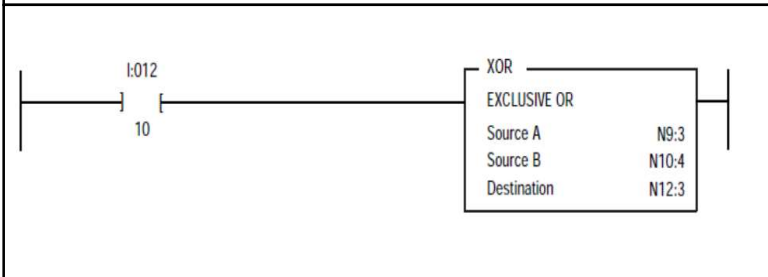
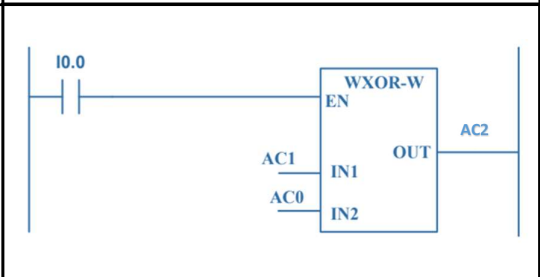
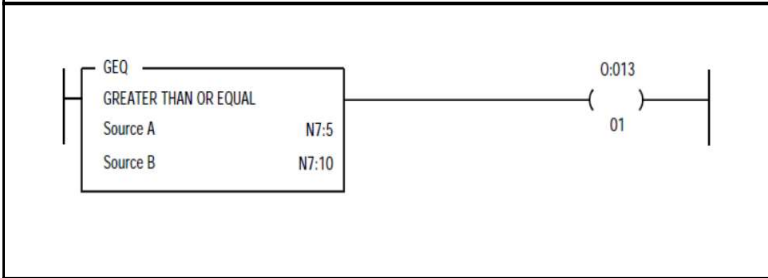
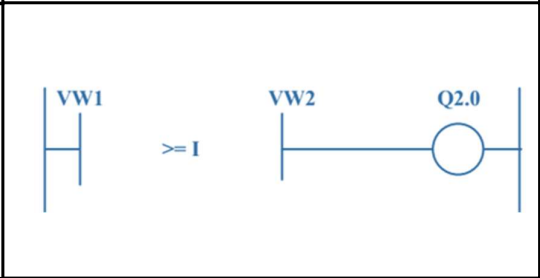
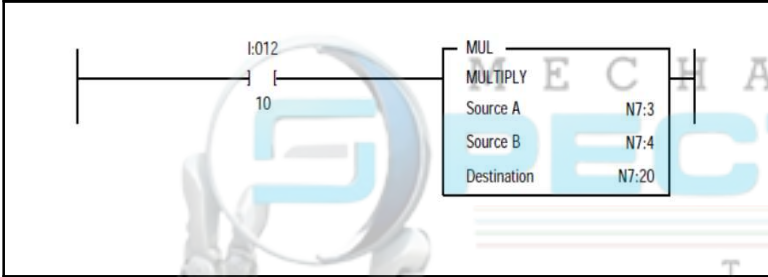
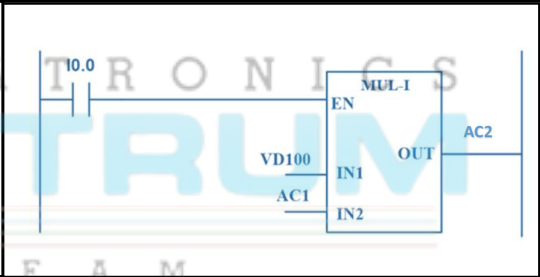
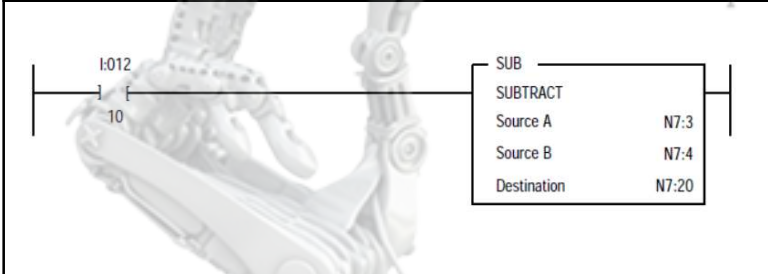
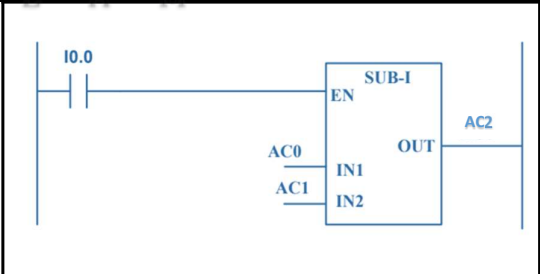


(20 Marks)

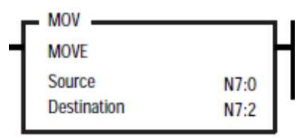
Draw ladder diagram using a Siemens S7-200 PLC for the following ladder diagrams of Allen-Bradly shown in the table. (using AC0, AC1, AC2, VD200, VD210, VD220, VW1, VW2, ... Siemens registers)

Allen Bradlly PLC Ladder diagram	Siemens PLC Ladder diagram
	
	
	
	
	
	

Allen Bradlly PLC Ladder diagram	Siemens PLC Ladder diagram
 XOR EXCLUSIVE OR Source A N9:3 Source B N10:4 Destination N12:3	 10.0 AC1 AC0 WXOR-W EN IN1 IN2 OUT AC2
 GEQ GREATER THAN OR EQUAL Source A N7:5 Source B N7:10 0:013 01	 VW1 VW2 Q2.0 Q2.0 Q2.0
 MUL MULTIPLY Source A N7:3 Source B N7:4 Destination N7:20	 10.0 VD100 AC1 MUL-I EN IN1 IN2 OUT AC2
 SUB SUBTRACT Source A N7:3 Source B N7:4 Destination N7:20	 10.0 AC0 AC1 SUB-I EN IN1 IN2 OUT AC2

Q2 (10 Marks)

State whether the following statements are true or false and record the answer in the table.

NO.	Statement	True or false
1.	In the following function, the N7:0 is the register that stored the data 	false
2.	A program is a user-developed series of instructions that directs the PLC to execute an action.	true
3.	The scan is normally a continuous and sequential process of reading the status of inputs, evaluating the control logic, and updating the outputs.	true
4.	And Load (AND LD) instruction is logically two branches of a rung in parallel	false
5.	In PLC operation cycle the controller examines the status input device, executes the user program, and changes the outputs accordingly.	true